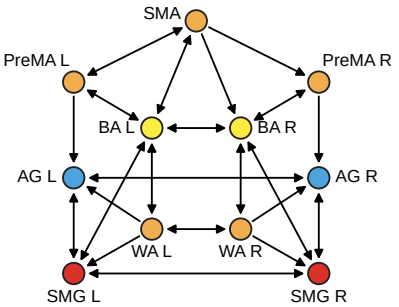
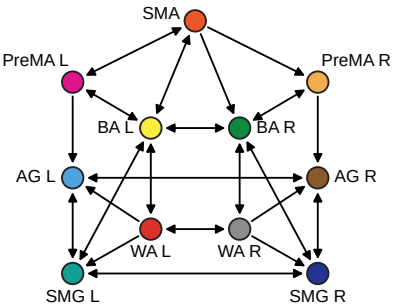
Symmetry analysis of inferred resting state structural network

a)

Fibration symmetry ≠ Group symmetry



4 Fibres



11 Orbits

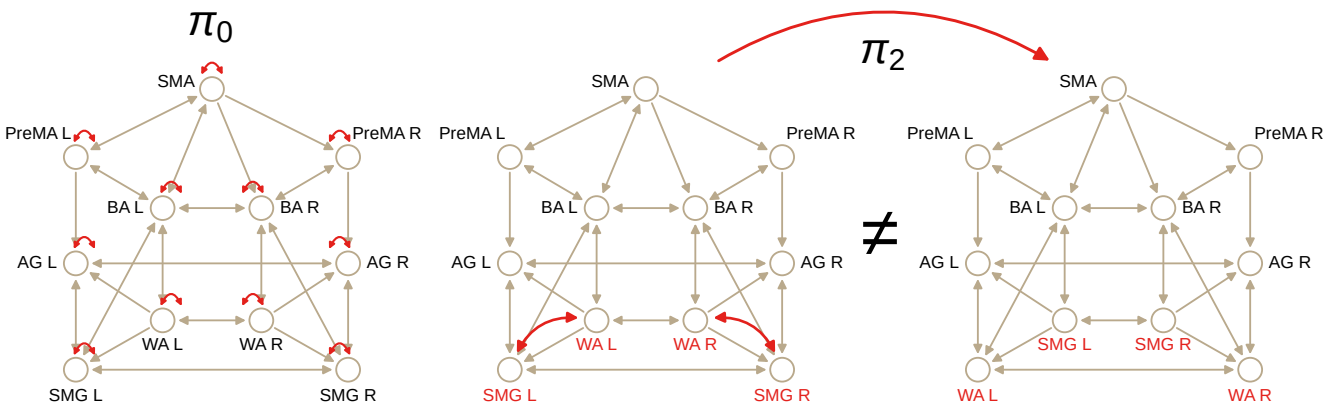
1: SMA
2: PreMA L
3: PreMA R
4: BA L
5: BA R
6: AG L
7: AG R
8: WA L
9: WA R
10: SMG L
11: SMG R

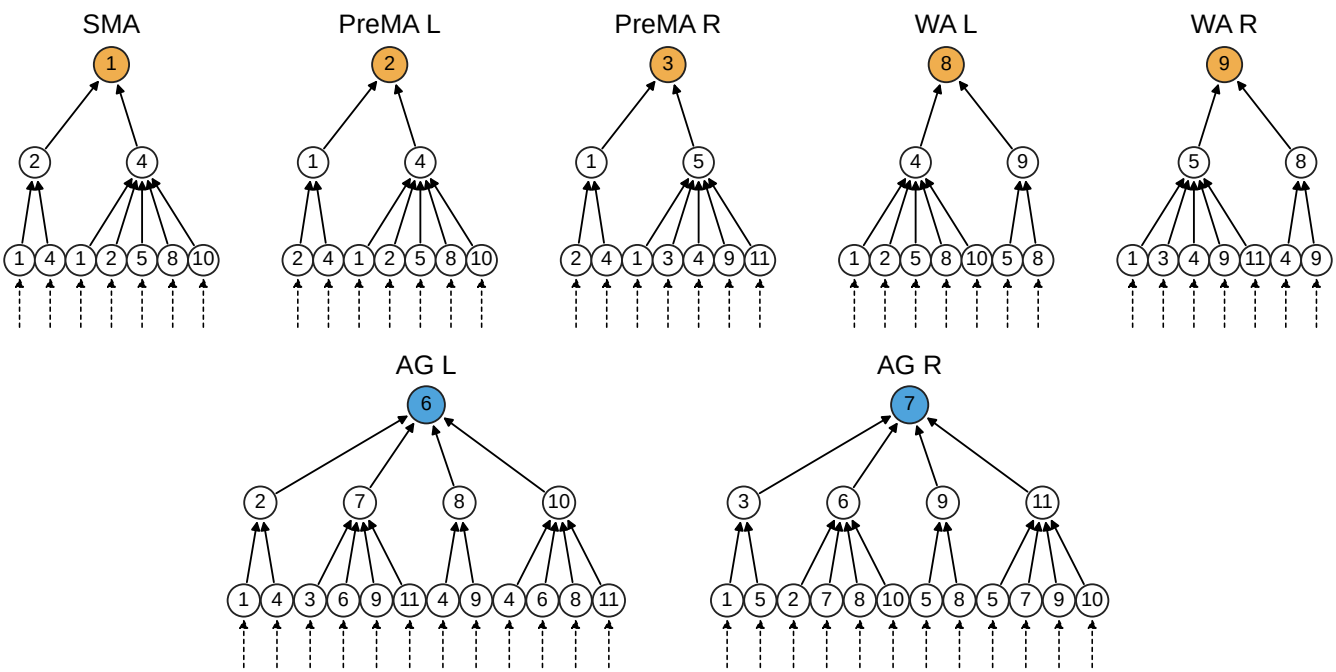
b)

Automorphisms

$$\pi_0 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{pmatrix}$$

$$\pi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 10 & 11 & 8 & 9 \end{pmatrix}$$



c)

Isomorphic Input Trees